MIL-P-46601B(MU)
14 June 1976
SUPERSEDING
MIL-P-46601A(MU)
30 June 1969

MILITARY SPECIFICATION

PRIMER, PERCUSSION: M36A2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers metal parts and the assembly of one type of percussion primer, M36A2, for use in 20mm cartridges.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Ammunition Lot Numbering

DRAWINGS

ARMAMENT COMMAND

LI 7259094 - Index of Inspection Equipment List for: Primer, Percussion, M36A2
D7259094 - Primer, Percussion, M36A2
D7259106 - Cartridge, 20mm, Target Practice, M204

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2.1 (Cont)

PUBLICATIONS

U. S. ARMY MATERIEL COMMAND

ORD-SIP-S314 - Visual Inspection Standards for Small Arms Ammunition Primer Defects
AMCR 715-505 - Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications.-The following publications, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

Code of Federal Regulations

49CFR 71-90 -- Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1967 Edition and revision) available from the Superintendent of Documents, Government Printing Office, Washington, D. C., 20315. Orders for the above publication, cite 49CFR 71-90 (latest revision)

3. REQUIREMENTS

3.1 General. -The primer shall be as specified on Drawing (Dwg.) D7259094, refe enced specifications and other requirements specified herein.

3.2 Manufacturing process.-All parts and assemblies shall be manufactured by a process approved by the contracting officer and no deviation from that process shall be made without his prior approval. (see 6.5)

3.3 Sensitivity.-The primer sensitivity shall be in accordance with the requirements of Dwg. D7259094.

3.4 Action time.-The action time of the test cartridge shall not exceed four milliseconds (ins) when assembled with the sample primers.

3.5 Function and casualty .-The sample primers shall function without casualty at ambient temperature when assembled into test cartridges.

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3.6 Workmanship. -The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and standards, and the following:

3.6.1 Process defects. -The primers and components shall be free of defects such as cracks, splits, severe scratches, cuts, dents, perforations, missing or unfinished components and other forms of substandard workmanship.

3.6.2 Cleanliness. -The primers and its components shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. -Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions .-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 Classification of Inspections. -The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 First article sample.-The sample shall be delivered for inspection in accordance with first article approval provisions of the contract and MIL-A-2550, and shall consist of:

a. Inspection:

125 cups
125 anvils

b. Firing tests:

2,000 primers

4.3.1.1 Inspection.-After inspection by the contractor and provisional acceptance by the Government activity, the sample will be inspected in accordance with the provisions of MIL-A-2550. Failure of the original first article sample to be in accordance with the applicable drawings and the requirements of Para. 3, as specified herein, will result in sample disapproval. (see 6.2)

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4.4 Quality Conformance Inspection.

4.4.1 Submission of product. -The product shall be submitted in accordance with MIL-STD-105, as applicable.

4.4.1.1 Lot.-A lot shall consist of:

a. Primers assembled by one manufacturer, in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision.

b. Metal parts produced by one manufacturer, in one unchanged process, in accordance with the same drawing and drawing revisions and same specification and specification revision.

c. One primer composition.

4.4.1.2 Lot identification.-Each lot of primers shall be identified in accordance with MIL-STD-1168. Each lot of packed primers shall be further identified with National Stock Number (NSN) and in accordance with 5.1 and 5.2.

4.4.2 Examination. -One hundred percent examination shall be performed by the contractor for all critical defects. Examination for major and minor defects shall be performed as specified herein, Each component and assembly found to be defective shall be rejected.

4.4.2.1 Sampling and acceptance criteria. -The sampling plans and procedures shall be in accordance with MIL-STD-105. To determine product acceptability, major or minor defects as listed herein may be considered collectively on a class basis or they may be considered individually. However, where major defects are considered collectively on a class basis, the acceptance number for any individual defect of the class shall be limited by the assigned Acceptable Quality Level (AQL) for individual defect associated with the class. (see 4.4.2.2.c) (see 6.3)

4.4.2.2 AQL's.-The following AQL's are assigned to major and minor defects:

a. Individual:

Major defects	0.25%
Minor defects	0.40%

b. Class basis:

Major defects	1.50%
Minor defects	2.50%

c. Individual defect associated with the class basis:

Major defects	0,40%

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4.4,3 Classification of defects.-The classification of defects shall be as specified in Table I.

TABLE I

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
1	No anvil	X			
2	Double anvil	X			
3	No charge	X			
4	No foil		X		
5	Double or multiple foil		X		
6	Slipped foil				X
7	Inverted or sidewise inserted anvil	X			
8	Crushed or mutilated				X
9	Loose composition				X
10	Deep seated anvil		X		
11	Cocked anvil				X
12	Low anvil		X		
13	High anvil				X
14	Excessive lacquer			X	
15	Loose anvil		X		
16	Stained or corroded metallic component(s)				X
17	Foreign matter		X		
	Cup				
18	Bell-shaped		X		
19	Bad bevel			X	
20	Brass shavings				X
21	Cracked				X
22	Crooked or eared				X
23	Dents				X
24	Crown on flat cup				X
25	Laminated metal				X
26	Ringed				X
27	Scratches				X
28	Nicked				X
29	V or U				X
30	Low cup			X	
	Anvil				
31	Burred			X	
32	Chipped				X
33	Filled		X		
34	Partial	X			

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4.4.3 (Cont)

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Anvil (Cont)				
35	No vent holes	X			
36	Laminated metal		X		
37	Ringed			X	
38	Unsymmetrical vent holes			X	
	Foil				
39	Partial			X	
40	Stained	X			
	Pellet				
41	Heavy		X		
42	Light	X			
	Gaging Primer				
43	Diameter		X		
44	Excess ellipticity between long and short axis diameters		X		
45	Height		X		

1/ Refer to ORD-SIP-S314 for visual defect standards for defects 1 through 40.

4.4.4 Tests.-Tests listed in Table II shall be performed on each primer lot sample in accordance with the test methods prescribed in 4.5. Sample size and acceptance criteria for each test shall be as specified. Only primers which have met the visual and dimensional requirements shall be used in the ballistics tests, selected in such a manner that the sample is representative of the entire lot. The primers selected shall be thoroughly mixed before being divided into samples for the various tests.

TABLE II

Test		Sample Size	Rqmt Para
Sensitivity	1/	400	3.3
Action time	2/	50	3.4
Function and casualty	3/	200	3.5

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4.4.4 (Cont)

1/ If the average drop height ($\bar{H}$) plus 5 standard deviations (5s) exceeds the maximum requirement or if $\bar{H}$ minus minus 2s is less than the minimum requirement, the lot shall be rejected subject to retest. If on retest the sample fails to meet the above criteria the lot shall be finally rejected,

2/ If the sample average action time plus 4 standard deviations exceeds 4 milliseconds or an individual primer action time of the test sample exceeds 4 milliseconds the lot shall be rejected.

3/ If 2 or more misfires, 2 or more pierced primers, or 10 or more primer leaks occur, the lot shall be rejected. If 1 misfire, or 1 pierced primer, or 4 or more primer leaks occur, a second sample consisting of double the quantity specified in Table 11 shall be fired. If in the cumulative sample (first test and retest) 2 or more misfires, 2 or more pierced primers, or 10 or more primer leaks occur, the primer lot shall be rejected.

4-.4.4.1 Reduced testing, function and casualty.-When the manufacturer of the primer is also the cartridge loader, the function and casualty test will be performed on first article sample only.

4.4.5 Packaging, Packing and Marking Examination. -Each lot of packaged primers shall be inspected for compliance with the requirements of 5.1 and 5.2.

4.4.6 Inspection equipment. -Index of Inspection Equipment List LI 7259094 identifies the applicable Inspection Equipment Lists required to perform examination and tests as specified herein. Equipment design(s) shall be in accordance with the applicable LI code designations. The code designations are defined on Dwg. B11075228, a detail of LI 7259094. The provisions of MIL-A-2550 shall apply.

4.5 Test methods and Procedures.

4.5.1 Sensitivity. -The methods of test shall be as specified in AMCR 715-505, Volume 8. The test technique shall be a complete rundown test using 50 sample primers at each height. All retests shall be conducted with complete rundown method using 100 primers at each height regardless of the method used on first test.

4.5.1.1 Reduced sensitivity testinR.-Reduced testing (two height test) may be instituted providing the criteria set forth in 6.4 are satisfied. However, when the two height test is being used, one randomly selected lot in ten shall be tested using the complete rundown test.

4.5.2 Action time. -The methods of test shall be as specified in AMCR 715-505, Volume 8. The test primers shall be loaded into test cartridges, 20mm, TP, M204, in accordance with Dwg. D7259106. Test cartridges shall be at ambient temperature.

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4.5.3 Function and casualty. -The methods of test shall be as specified. in AMCR 715-505, Volume 8. The test sample primers shall be loaded in test cartridges; 20mm, TP, M204, in accordance with Dwg. D7259106 and fired in bursts of 50 cartridges. The test cartridges shall be at ambient temperature. The weapon barrel shall be at ambient temperature at beginning of test and cooled to ambient after each burst. The cyclic rate of each burst shall not be less than that of the weapon requirement of the test cartridge detailed specification. The cyclic rate for each burst shall be recorded for informational purposes.

4.5.4 Defective weapon.-If the test weapon fails to comply with the minimum cyclic rate established for the weapon, or any test cartridge firing defect is found to have been caused by a defective test weapon, the test shall be suspended. The weapon shall be corrected or replaced and the test repeated in whole or in part, as indicated. If the test cartridge defect is not found to have been caused by the defective weapon, it shall be valid for lot penalty in accordance with Table II.

5. PREPARATION FOR DELIVERY

5.1 Packaging and Packing, Level C. (Immediate use). -Where packing is required, the primers shall be packed in containers so as to assure acceptance by common or other carrier for safe transportation, in accordance with CFR Title 49 71-90 (ICC Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles), at the lowest rate, to the point of delivery. The primers shall be packed and spaced in such a manner that a barrier of appropriate material is present between the crown of one primer and the base of another primer to resist mass detonation and shield the primers from contact with flame and shock. The primers shall be packed in such a manner as to be protected from moisture.

5.2 Marking. -Where packing of primers is required, the packing boxes shall be marked in accordance with ICC rules and regulations and the marking shall identify the supplier, contract number, item, lot number, drawing number and revision, 'primer mix number, month and year of manufacture, the primer quantity, the box cube and gross weight.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Place of inspection and tests if other than place of manufacture.

6.1.3 Provision for the submission and approval of the manufacturing process. (see 3.2)

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6.1.4 Instructions for shipment of first article and production samples for firing tests.

6,1.5 Provision for furnishing the National Stock Number (NSN) if packing and marking for shipment is required. (see 5.1 and 5.2)

6.1.6 Provision for the supply, maintenance and disposition of Government furnished equipment for acceptance inspection purposes.

6.1.7 Provisions for the inclusion of MIL-STD-1167, Ammunition Data Cards on DD form 1423, Contract Data Requirement List.

6.2 First article sample. -In order to minimize inspection costs, the firing test, as specified in 4.3, will be performed after the sample has been found acceptable for all other requirements.

6.3 AQL's.-The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitations, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

6.4 Sensitivity tests. -The criteria for use of the two height method and for reinstitution of the rundown method is specified in 6.4.1 and 6.4.2.

6.4.1 Rundown to two heights. -When the complete rundown test is in effect, the two height test shall be instituted providing this two height method is considered desirable by the Government and all of the following conditions are satisfied:

a. The preceding five lots have been accepted by the rundown method on original test.

b. Skewness value for each of these lots has not exceeded the value of .787 (without regard to its plus or minus sign).

c. Production is at a steady rate.

6.4.2 Two height to rundown. -When the two height test is in effect, the rundown method shall be instituted if any of the following occur:

a. A lot is rejected, first test,

b. Skewness value exceeds the value of .787 (without regard to its plus or minus sign).

c. Production becomes irregular or delayed.

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6.5 Process deviation.-A process deviation is defined as a change in the approved basic method of manufacture or an operational change which may alter the metallurgical, chemical or physical properties of the item.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-P-46601B(MU) Primer, Percussion: M36A2

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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